

Necrotizing soft-tissue infections and primary sepsis caused by *Vibrio vulnificus* and *Vibrio cholerae* non-O1.

BACKGROUND: *Vibrio* species are a rare cause of necrotizing soft-tissue infections and primary septicemia, which are likely to occur in patients with hepatic disease, diabetes, adrenal insufficiency, and immunocompromised conditions. These organisms thrive in warm seawater and are often present in raw oysters, shellfish, and other seafood. This study examined fulminating clinical characteristics of *Vibrio vulnificus* and *Vibrio cholerae* non-O1 soft-tissue infections and identified outcome predictors.

MATERIALS: Thirty patients with necrotizing fasciitis and sepsis caused by *Vibrio* species were retrospectively reviewed. Twenty-eight patients had a history of contact with seawater or raw seafood. Eight patients had hepatic disease such as hepatitis or liver cirrhosis, and seven patients had diabetes mellitus. Nine patients had hepatic dysfunction combined with diabetes mellitus. Microbiology laboratory culture studies confirmed *V. vulnificus* in 23 patients and *V. cholerae* non-O1 in seven patients.

RESULTS: Surgical debridement or immediate limb amputation was initially performed in all patients with necrotizing soft-tissue infections. Eleven patients (37%) died within several days of admission and 19 survived. The mortality of *V. cholerae* non-O1 group (57%) is higher than that of the *V. vulnificus* group (30%). A significantly higher mortality rate was noted in patients with initial presentations of a systolic blood pressure of $< \text{ or } = 90$ mm Hg, leukopenia, decreased platelet counts, and a combination of hepatic dysfunction and diabetes mellitus.

CONCLUSIONS: *Vibrio* necrotizing soft-tissue infections should be suspected in patients with appropriate clinical findings and history of contact with seawater or seafood. *V. cholerae* non-O1 may cause bacteremia more often than *V. vulnificus* in patients with liver cirrhosis. Early fasciotomy and culture-directed antimicrobial therapy are aggressively recommended in patients with hypotensive shock, leukopenia, high band forms of white blood cells, decreased platelet counts, severe hypoalbuminemia, and underlying chronic illness, such as hepatic dysfunction and diabetes mellitus.